

Clinical Risk Assessment Report: Patient 94

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 45.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.07x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +4.0%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.31 [Bottom 22%] (+4.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.76 [Top 22%] (+3.8%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.57 [Avg (73th %ile)] (+3.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.76 [Avg (73th %ile)] (+3.4%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.65 [Avg (66th %ile)] (+3.3%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Patient Age (Age) **Value: 35.0 [Bottom <1%] (+2.8%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.46 [Avg (72th %ile)] (+2.4%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

25th Percentile Metabolic Activity (CONVENTIONAL_SUVbwQ1) **Value: 0.71 [Top 17%] (+1.6%)**
Reflects the baseline metabolic activity and glucose consumption of the tumor tissue.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

